

Grief-Stricken in a Crowd: The Language of Bereavement and Distress in Social Media

Jed R. Brubaker¹, Funda Kivran-Swaine², Lee Taber¹, and Gillian R. Hayes¹

¹School of Information and Computer Sciences
University of California, Irvine
Irvine, CA, USA
{jed.brubaker, ltaber, hayesg}@uci.edu

²School of Communication and Information
Rutgers University
New Brunswick, NJ, USA
funda@rutgers.edu

Abstract

People turn to social media to express their emotions surrounding major life events. Death of a loved one is one scenario in which people share their feelings in the semi-public space of social networking sites. In this paper, we present the results of a two-part investigation of grief and distress in the context of messages posted to the profiles of deceased MySpace users. We present coding system for identifying emotion distressed content, followed by a detailed analysis of language use that lays a foundation for natural language processing (NLP) tasks, such as automatic detection of bereavement-related distress. Our findings suggest that in addition to words bearing positive or negative sentiment, linguistic style can be an indicator of messages that demonstrate distress in the space of post-mortem social media content. These results contribute to research in computational linguistics by identifying linguistic features that can be used for automatic classification as well as to research on death and bereavement by enumerating attributes of distressed self-expression in a post-mortem context.

Introduction

Social media-based interactions are an increasingly important to many people's social lives. Life's significant events, from birthdays to marriage proposals, are now experienced in part via social network sites (SNS). This remains true of when those lives come to an end. Grieving a loved one's death—once limited to the physical world—now extends to the social networking profiles of the deceased. Following a user's death, friends often express their shock and grief on the deceased's profile page (Brubaker & Vertesi 2010; Getty *et al.* 2011). Over time, survivors continue to return to social media and keep those who have passed close through continued updates on events and feelings that would have been shared with the deceased were they alive (Brubaker & Hayes 2011).

Analyzing content from sites such as MySpace provides insight into the role that social media plays in contemporary experiences of death, as well as the impact of death on these digital spaces. So far, much of the previous work in the area of post-mortem interactions in social media has been qualitative in approach, employing content and thematic analyses to understand grieving practices. However, quantitative analyses of the content and linguistic style of post-mortem comments are crucial in understanding practices during bereavement. Moreover, user-generated content posted via social media provides a unique opportunity to observe grieving practices in a naturalistic setting. Messages in SNS are in their intended social environments as opposed to interview or lab settings, where expressions of grief have historically been studied.

This work focuses on extreme expressions of grief and mourning in SNS following the death of a friend or loved one. In the first part of this paper we present a codebook for identifying distressed comments based on a thorough review of the literature on grieving practices and bereavement. This coding scheme identifies comments exhibiting emotional distress as a potential indicator of unhealthy mourning and/or the need for additional support. Based on comments coded through this scheme, in the second part of this paper we present the results of a detailed linguistic analysis of comments posted to deceased MySpace profiles in order build explanatory models based on linguistic features that contrast comments exhibiting emotional distress with those that display more conventional funerary language. To this end, we examined 2213 post-mortem comments posted to the profiles of 652 MySpace users following their deaths.

The detailed linguistic analysis of distressed comments presented here expands our current knowledge around the use of language in online grieving. Existing research in this space has predominantly focused on the benefits of such spaces that enable individuals to express their grief. A review of content, however, reveals that some of the

bereaved could benefit from additional social or clinical support. Likewise, the continued presence of the dead amidst lists of friends appears to result in prolonged distress (Brubaker & Hayes 2010). Our code system identifies MySpace comments demonstrating such distress while our computational investigation of language indicates which linguistic features are most informative for identifying emotional distress in this context. As such, this work contributes to enabling future research in natural language processing (such as automatic detection of emotional distress in social media). Detection of comments expressing emotional distress can be beneficial for directing appropriate support during times of social and psychological hardship.

Theoretical Background

Death in the context of social media

Memorializing the dead online has been a practice since the early days of the Internet. Initially memorials consisted of individually maintained webpages and later “virtual cemeteries” hosting “cyber memorials” that allowed visitors to post messages (Roberts & Vidal 2000). With the rise of social media, users also started reappropriating the profile pages of deceased users (Brubaker & Hayes 2010) and creating SNS groups to memorialize the deceased (DeGroot 2008; Marwick & Ellison unpublished).

As Brubaker & Hayes (2011) note, however, SNS profile pages are distinct from other memorial spaces as they are created by the deceased while they are alive, and not by friends or family post-mortem. Following a user’s death, profiles are flooded with comments posted by friends of the deceased, expressing shock and communicating their remorse. A number of scholars have noted that commenters rarely interact with each other on the deceased’s profile page and instead speak directly to the deceased (DeGroot 2008; Dobler 2009). Brubaker & Hayes (2011) argue that this tendency, in conjunction with the continued presence of deceased friends and their profile pages in SNS, provides the ability for the bereaved to engage in “post-mortem social networking” as a way of maintaining connections with the deceased.

In a classification of 200 MySpace comments, Carroll & Landry (2010) found similar patterns showing that users turn to post-mortem profiles to write expressions of grief and admiration, as well as share memories of the deceased. Meanwhile, Getty *et al.* (2011) performed a linguistic study of the wall posts of memorialized Facebook profiles and found that that post-mortem Wall posts show higher rates of negative emotion than pre-mortem equivalents.

Across the existing research, the ability for the bereaved to visit post-mortem profiles is generally regarded

positively as it provides a space for “disenfranchised grievers” (Doka 1989) to mourn, such as friends (Skylar & Hartley 1990) or others who are not family or members of the immediate community. However, these spaces can also be sources of pain. When discussing comments that serve to maintain connections with the deceased, Brubaker & Hayes (2010) noted that some authors repeatedly post distressed comments over long periods of time. The continued presence of the deceased in an individual’s social network may have ramifications given that the survivor must make the active choice to remove the deceased from their network (Brubaker, Hayes, & Dourish in press).

Grief & Mourning

While there are many definitions of both grief and mourning (*e.g.*, Kastenbaum 1979; Klass *et al.* 1996; Kubler-Ross 2005), Rando (1993) provides a useful delineation between these two terms with definitions that focus on the potential psychological states and complications experienced by the bereaved. *Mourning* refers to the overall process through which an individual comes to acknowledge and accommodate the loss, while *acute grief reactions* are defined as psychological, behavioral, social, and physical reactions to the perception of that loss. Thus, while mourning is an ongoing process, posting a comment to a friend’s MySpace page is an acute event serving as a manifestation of grief.

Current approaches to death and dying generally agree that the bereaved do not follow a uniform or finite grieving process. As such, there is no ideal duration for mourning or way in which to grieve. Instead, the objective of mourning is for the bereaved to incorporate the reality of the death into their relationship with, and understanding of, the deceased (Rando 1993). This objective is held while acknowledging that the bereaved often continue to have bonds with the deceased (Klass *et al.* 1996).

To this end, researchers and clinicians have created a variety of frameworks that enumerate objectives for the bereaved. Rando, for example, outlines six requirements for healthy mourning: recognition of the loss; reaction to the separation; recollections of the deceased; relinquishing of attachments to the deceased; readjustment to the world without the deceased; and a reinvestment of energy into other people, activities, etc. In Rando’s framework, these requirements assist the bereaved in reorienting themselves to a reality that incorporates their loss.

Studying the writing of the bereaved

Previous research has found that writing to and about the deceased allows the bereaved to express their grief and gain perspective on their loss (see Roberts & Vidal 2000 for review). Emotional disclosure via writing following the

love of a loved one can have positive effects, such as successful coping (Cable 1996). Translating traumatic events such as death into written language can help individuals encode and structure their experiences in an organized way (Pennebaker *et al.* 1997). These benefits are particularly apparent when the death is sudden or tragic (Roberts & Vidal 2000) as if often the case with social media where, until recently, users have been young. However, as researchers have had little access to personal writing, these claims tend to be based on historical journals, content produced for public consumption, and self-reflection and survey studies (Roberts & Vidal 2000). In contrast, SNS provide a naturalistic setting to observe the writing of the bereaved. Messages in SNS are in their intended social environments as opposed to interview or lab settings, where expressions of grief were most frequently studied in past.

Data

The data used for the work we present in this paper are a subset of a larger dataset of comments collected from profiles of 1369 deceased MySpace users in April 2010. Deceased profiles were identified using MyDeathSpace (MDS)¹, a website dedicated to connecting obituaries and/or news of deaths to existing MySpace profiles. MDS contains more than 15,000 user-submitted entries, as well as comments on individual entries, including additional information and links to other online content (*e.g.*, newspaper articles). The deceased status of each profile owner was verified by one of the researchers based on comments posted by friends or profile content.

For this study, we limited our sample to comments posted to profiles of users who lived in the United States and who had been dead for at least three years. Additionally, only comments from publicly visible profiles were collected. Publicly visible profiles made up the majority of profiles given the lack of privacy features on MySpace during the timeframe in which profile owners died.

Finally, two types of profiles that are substantially different from typical profiles were omitted from our sample: those belonging to celebrities (*e.g.*, Elvis) and profiles that had been created as or repurposed into “issue-profiles” by survivors of the deceased (*e.g.*, those focused on issues such as substance abuse or war rather than on a specific individual who had died from causes related to those issues).

The subset used for this study consists of 2213 randomly selected comments from a total dataset of 206,068 comments. Comments in this subset were posted to 652

profiles by 1660 unique comment authors. The deceased, on average, died at age 27.68 (SD=13.06) and 66.7% of the deceased users were male.

Coding for Emotional Distress

We generated a codebook for manual coding of comments exhibiting emotional distress on post-mortem MySpace profiles following a review of literature on expressions of grief and sorrow during mourning and bereavement. The development of the codebook began with an acknowledgement that much of the language in post-mortem comments is performative and uses formulaic language. Thus, we distinguished between common funerary sentiments (*e.g.*, “I miss you already...”) and those expressing what we ultimately labeled as emotional distress (ED).

Two authors initially identified twenty extreme comments that unambiguously were written by an author expressing distress, such as the following:

i cant stop thinking about u lately .. its been crazy especially friday considering it was ur 100 day.. ppl keep telling me to stop counting the days but i wont.. its the only way i feel close to you ... ah i've been so confused lately over everything even the smallest things.. like the whole time i was i the hospital alls i could do was think of you nd lately thats all i can do.. its crazy er i hate this so much i miss you like crazy u were so easy to talk to its like no matter wut u always understood.. idk w.e. i just needed to "talk" to you i guess.. love you babii

Using a grounded approach, we open coded comments to identify specific features that distinguished them from non-ED comments. These features were then used to develop tentative rules that specify what does and does not qualify a comment as ED. For example, “Sharing Memories of the Deceased” was a common feature of ED comments, but not unique to them, resulting in rule #3: “*Memories of the deceased in and of themselves do not constitute ED.*”

Codebook Rules

The codebook started by defining ED as negative emotions, specifically pain and sadness associated with the deceased. Language associated with ED involves words like hurt, pain, sorrow, and metaphors related to these words (*e.g.*, “my heart has been broken”, “my life has been shattered”, etc.).

The codebook included six rules to aid in distinguishing between comments, each including example comments that did and did not qualify as ED based on that rule (see Table 1 for summary). The codebook was conservative in attributing ED to comments, with Rule #1 establishing

¹ www.mydeathspace.com